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LECTURE 1 · INTRODUCTION

Mobile Application and Service Development

Razvoj mobilnih aplikacija i servisa

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Faculty of Electrical Engineering, University of Tuzla · 1 October 2026

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Intro survey: 5 minutes

Tell me who you are, what you use and how you use AI.

We will look at the AI answers together, live.

Your answers do not affect your grade.



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Last year's final projects

While you watch, keep one question in mind:

> What could go wrong when a real user installs this?

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Why mobile is different

Battery

Every sensor, request and animation costs energy.

Network

Connections drop, slow down and switch between Wi-Fi and mobile data.

Permissions

Users can say no to camera, location or notifications.

Fragmentation

Thousands of devices, screen sizes and Android versions.

Lifecycle

The system can pause or kill your app at any moment.

Every one of these is something AI-generated code usually ignores.

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AI can generate an Android app in minutes.

This course is about everything that happens after: when it crashes on a real phone, drains the battery, or leaks an API key.

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Who we are

LECTURER

dr. sci. Alma Šećerbegović, van. prof.

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Consultations: [Wednesday 14:00–16:00]

TEACHING ASSISTANT

Lejla Šarić, stručnjak iz prakse

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Consultations: [to be announced]

Slides in English · lectures in Bosnian · consultations in both languages

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What you will be able to do

Build solutions for modern mobile platforms, and understand what makes mobile services unique given network and device characteristics.

01

Recognize the capabilities and limitations of mobile devices.

02

Design apps with multiple screens and system components.

03

Develop apps on a popular mobile platform: Android.

04

Evaluate app performance against defined criteria.

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How you are graded

10

Activity

Attendance and participation in lectures, labs and workshops

20

Tests

Test 1 mid-semester (10) and Test 2 at the end (10)

25

Homeworks

3 homeworks, each defended live in the lab

45

Final project

Built in 3 phases, defended in exam week

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Three homeworks

#	Topic	Points	AI	Published → Defended
1	Kotlin basics: collections, null safety, classes	7	Not allowed	Week 2 → Week 4
2	Compose UI and state: rebuild a screen from a screenshot	8	Allowed	Week 4 → Week 6
3	Bug hunt: find, fix and explain bugs in networking and lifecycle code	10	Allowed	Week 11 → Week 13

How defenses work

A 10-minute live change to your own code, without AI. If you understand your code, this is easy.

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The final project: one app, three phases

5 pts · Week 7

Phase 1

Idea, mockups of all screens, navigation sketch, empty Android Studio project

15 pts · Week 10

Phase 2

Real Compose screens, working navigation, data from an API or database

25 pts · Exam week

Phase 3

Complete app, polished UI, extras, README and live defense

To pass the project, you need at least 10 of the 20 points from Phases 1 and 2.

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What every project needs

Core

- At least 3 meaningful screens
- Navigation with correct back behavior
- State survives rotation and interruptions
- Material 3, Lazy lists, accessibility

Extras: at least 2

- Location, camera, sensors or audio
- Public API, local database or cloud
- Files, DataStore or sharing
- Used meaningfully, not just shown

New this year

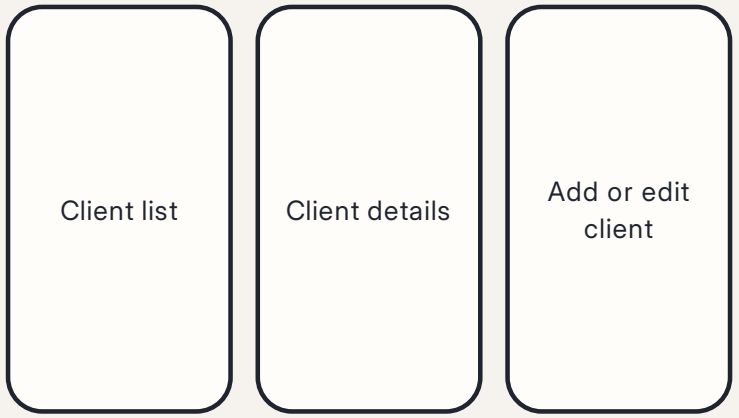
- Tested on a real Android phone
- One measured metric: startup time, memory or offline behavior
- README section: where AI was wrong and how you found it

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Minimum example: Hair Salon Manager

A hairdresser keeps client records: name, visit date, hair color used and notes. Goal: never forget what was used on whom.

This is the minimum acceptable project. Yours should be more complex and more original.



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Weekly plan: weeks 1–8

Week	Lecture	Lab and deadlines
1	Introduction · why mobile is different	Install Android Studio, no lab
2	Kotlin essentials: reading, writing and reviewing code	Homework 1 published
3	No lecture (workshop)	Kotlin practice
4	First Compose app, layouts and components	Lab: first Compose app
5	State and recomposition	Lab: Homework 1 defended. Homework 2 published.
6	Lists and navigation	Lab: lists and navigation.
7	Lifecycle and ViewModel	Lab: Homework 2 defended
8	Test 1 + Phase 1 presentations	

This might change!

Weekly plan: weeks 9–15

Week	Lecture	Lab and deadlines
9	Coroutines and networking · keeping API keys safe	Lab: API call.
10	Local data: Room, DataStore, offline-first	Lab: Room.
11	Repository pattern and background work	Lab: Phase 2 check. Homework 3 published.
12	New: sensors, camera, location, permissions	Sensors lab
13	New: testing and performance on real devices	Homework 3 defense
14	Holiday	—
15	Test 2	Final project defense in exam week

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AI in this course

AI is allowed for building. Understanding is what gets graded.

Allowed

- Project work
- Homeworks 2 and 3
- Explaining errors and debugging
- Writing tests

Not allowed

- Homework 1
- Tests
- All live defenses

Always required

- Name the tool and prompt in comments and README
- Explain AI code in your own words
- Same for Stack Overflow; other students' code is never allowed

Unreferenced, unexplained AI code can cost you all points for that task.

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What you need

Tools

- Laptop, 16 GB RAM recommended
- GitHub account
- Android Studio, latest stable
- An Android phone for testing (recommended)

Communication

- Google Classroom: materials and deadlines
- Questions: Classroom stream or private comments
- Email for personal matters

Literature

- Lecture slides (required)
- Android Programming: The Big Nerd Ranch Guide, 5th ed.
- developer.android.com/courses

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Before the first lab

1

Install Android Studio

Latest stable version. Create and run a Hello World project.

2

Set up GitHub

Create an account if you don't have one.

3

GitHub Student Pack

Apply for GitHub Student Pack if you don’t have one.

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Questions?

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